Quinten Pienaar | +1 781 518-8110 | quinten.pienaar@gmail.com | https://github.com/qpienaar/school_projects | http://www.linkedin.com/in/qpienaar |

Education

Rensselaer Polytechnic Institute, Troy, NY
Masters of Science in Mechanical Engineering, GPA 3.92 Spring 2026-Fall 2026
Bachelors of Science in Mechanical Engineering, Minor in Economics, GPA 3.53, Cum Laude Fall 2022-Fall 2025

Professional Interests

Interested in research-oriented engineering roles focused on advancing computational modeling and simulation, high-performance computing, robotics, controls, and machine learning.

Experience

Masters Thesis Center for Scientific Computation at Rensselaer Polytechnic Institute, Troy, NY Summer 2025-Present

- Developed software to convert nuclear reactor models from Constructive Solid Geometry to Boundary Representation, supporting both fusion and fission models.
- Generated unstructured tetrahedral meshes of reactor models to evaluate mesh-based neutronics and coupled simulation methods.
- Conducted coupled FEM heat conduction simulations across mesh resolutions, evaluating computational cost, accuracy, and convergence.
- Authored and defended M.S. thesis before faculty committee, presenting geometry conversion, simulation methodology, and research findings.

Product Security Intern, GE Vernova - Wind, Schenectady, NY Summer 2025

- Assessed industrial network vulnerabilities, proposing network segmentation and micropatching solutions for critical systems.
- Developed role-based cybersecurity training and evaluated secure-coding tools to strengthen internal development practices.

Control Systems Intern, GE Vernova - Gas Power, Schenectady, NY Summer 2024

- Tested, and qualified industrial control and digital networking devices for gas turbine customer environments.
- Simulated customer network configurations, integrated controls hardware/software

Projects

Cable-in-Conduit Thermal Analysis - Introduction to Computational Fluid Dynamics, Rensselaer Polytechnic Institute Spring 2026

- Developed CAD models of Cable-in-Conduit (CIC) cooling systems for superconducting magnets in fusion reactors.
- Applied unstructured meshes and thermal-fluid boundary conditions, before conducting high-fidelity coupled thermal-hydraulic simulations of CIC systems. Demonstrated CIC cooling systems ability to maintain operating conditions under thermal loads.

Robotic Path Planning with Neural Networks - Scientific Machine Learning Final Project, Rensselaer Polytechnic Institute Spring 2026

- Developed and evaluated neural network-based heuristics for robotic navigation, testing whether neural networks could improve upon Euclidean distance heuristics while maintaining efficiency across varied obstacle environments.
- Compared neural networks trained with mean squared error vs asymmetric mean absolute error loss functions and presented results to faculty.

Self Balancing Segway - Mechatronics Project, Rensselaer Polytechnic Institute Fall 2025

- Developed linearized state-space and nonlinear dynamic models of a two-wheeled Segway; designed an LQR controller in MATLAB to regulate position and maintain balance.
- Validated closed-loop control through simulation and hardware testing, demonstrating stable balancing and position regulation with disturbances.

HPC Parallel Image Recreation and Rendering - Parallel Computing Final Project, Rensselaer Polytechnic Institute Spring 2026

- Collaborated with a team of graduate students to develop a massively parallel, GPU-accelerated genetic algorithm for image reconstruction.
- Deployed the algorithm on RPI's AiMOS supercomputer and evaluated strong/weak scaling across multi-GPU and multi-node MPI configurations.
- Analyzed performance results and co-authored a conference-style paper documenting the methodology, scalability, and findings.

UAV Wing Spar Optimization Under Uncertainty - Numerical Design Optimization Project, Rensselaer Polytechnic Institute Fall 2025

- Optimized a carbon-fiber UAV wing spar under uncertain aerodynamic loading using MATLAB fmincon, surrogate modeling, and Latin hypercube sampling.
- Reduced spar mass by 70.68% while satisfying six-sigma probabilistic stress constraints under complex aerodynamic loading.

Leadership and Activities

Pi Lambda Phi Fraternity - President Fall 2024 - Fall 2025

- Led a seven-member executive board and 45-member chapter. Oversaw finance, housing, recruitment, risk management, alumni relations, fundraising, and member development.
- Standardized leadership transitions and position expectations through documented procedures, mandatory transition meetings, and performance incentives. Oversaw ~$20K in facility upgrades and improvements.

Rensselaer Mens Rugby - Captain, Head Coach Spring 2025 - Fall 2025, Fall 2027

- Led three weekly practices, team meetings, and film sessions while establishing team goals, expectations, and development plans.
- Led the team to its first National Collegiate Rugby 7s appearance and a Liberty League 15s runner-up finish.

Skills

Languages & Tools: Python, MATLAB, C++, C, CUDA, MPI, PyTorch, ROS2, Linux, Simulink, Siemens NX, Altair CFD, Slurm, FreeCAD, Gmsh
Engineering: CFD, FEM/FEA, HPC, Unstructured Meshing, GPU Computing, Machine Learning, Surrogate Modeling, Numerical Optimization, Control Systems, Robotics, Pneumatics, Mechatronics